Q.24 What are the different periodical check for aircraft maintenance?

Q.25 How does brake system work in a landing gear?

Q.26 What is fail safe concept?

Q.27 How the testing of fuel tanks is done?

Q.28 Name all the secondary control surfaces.

Q.29 What is stressed construction?

Q.30 Write a precautionary measure taken to avoid engine failure.

Q.31 Write hort note on utility of emergency exits.

Q.32 What are the different types of wings planforms?

Q.33 Write a short note on using Plywood in Aircraft Structure.

Q.34 What are the types of control system used in aircrafts?

Q.35 How are compositesmade?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Draw a family tree of aircrafts and classify the different types of aircrafts in detail

Q.37 What are the minor defects occuring on aircrafts? Describe the investigation process and the rectifications procedure of the minor defects.

Q.38 What is the use of composites on aircrafts? How the composites are made and what are their properties?

(20)

Roll No. ....

4th Sem / AME

Subject:- General Airframe and Aero Modelling

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Most of the transport aircraft have?

a) Dihedral wings b) Anhedral wings

c) Swept forward wings

d) Elliptical wings

Q.2 What is the function of trim tab control?

a) To minutely maneuver the a/c

b) Release pilot of stick force

c) To provide stability

d) All of the above

Q.3 What type of aircraft is primarily designed for transporting passengers over long distances?

a) Fighter jet b) Cargo plane

c) Commercial airliner d) Helicopter

Q.4 What type of load does the landing gear of an aircraft primarily support during take-off and landing?

a) Aerodynamic load b) Torsional load

c) Structural load d) Impact load

Q.5 What is Alclad?

a) Pure Aluminum coating to prevent corrosion
b) Aluminum and cadmium alloy
c) Whole surface cldded
d) None of the above

Q.6 What is the full form of STOL?

a) Short take-off and landing
b) Straight take-off and landing
c) Swift take-off and landing
d) None of the above

Q.7 Which load is associated with the weight of the aircraft itself and the payload it carries?

a) Aerodynamic load b) Structural load
c) Torsional load d) Dynamic load

Q.8 What is the purpose of secondary control system in aircraft?

a) To assist in controlling the aircraft during emergencies.
b) To provide backup control in case of primary system failure
c) To enhance the performance of primary control systems
d) To control non-essential functions such as lighting and entertainment systems

Q.9 Which of the following materials is commonly used for minor structural repairs in metal aircraft?

a) Carbon fibre b) Fiberglass
c) Aluminium d) Kevlar

Q.10 Which of the following is not true about an airframe made with carbon-fiber composite?

a) Decreases drag b) Decreases thrust
c) Higher cabin pressurization
d) Higher wing aspect ratio

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is the direction of lift in airplane?

Q.12 What do you mean pure monocoque?

Q.13 What are the different aircraft types?

Q.14 What are the benefits of using composite in an aircraft?

Q.15 How the brakes are applied on wheels?

Q.16 What is the importance of minor structural repairs in aircraft?

Q.17 How the wings ate attached to fuselage?

Q.18 What do you mean by fail safe concept?

Q.19 Give examples of tubular structure.

Q.20 How is duplicate inspection done?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Describe the necessity of emergency exits on the aircrafts.

Q.22 Discuss different types of structural arrangements.

Q.23 What are the types of secondary controls used?